How Pruning Criteria Shape Post-RLVR Recovery: A Controlled Study of Structured MLP Pruning in Reasoning LLMs

Anonymous ACL submission

Abstract

Fine-grained structured pruning of reasoning Large Language Models (LLMs), which removes MLP channel groups rather than entire layers, is typically followed by recovery training (SFT or Reinforcement Learning with Verifiable Rewards, RLVR) to restore reasoning capabilities. RLVR is widely believed to be superior, yet we show this hierarchy reflects suboptimal pruning, not a fundamental property of RL. When pruning preserves representational diversity, recovery method choice becomes irrelevant, a phenomenon we term *reward-accuracy decoupling*: Group Relative Policy Optimization (GRPO) and SFT produce near-identical accuracy despite vastly different reward trajectories, a pattern that holds at moderate sparsity (20–25%) across benchmarks. We propose Recovery-Aware Structured Pruning (RASP), which balances gradient importance with Centered Kernel Alignment (CKA)-based representational diversity via a coefficient α. On DeepSeek-R1-Distill-Qwen-7B at 20% sparsity, a dose-response analysis reveals an inverted-U in recovery accuracy, peaking at α=2.0 (66.11% on GSM8K). Under GRPO training, four pruning criteria produce a *recovery capacity spectrum*: diversity-preserving pruning achieves active learning (peak reward 4.4× vs importance-only), while importance-dominated pruning collapses entirely. The spectrum tracks preserved representational diversity rather than gradient importance, with preliminary evidence at 1.5B scale and, via SFT-based recovery, across Qwen2.5-family models (Qwen2.5-Math-7B).

1 Introduction

Reasoning Large Language Models can be compressed through structured pruning, but the pruned model must recover its reasoning capabilities through fine-tuning. The dominant assumption is that the recovery paradigm (SFT versus RLVR) determines recovery quality: Ji et al. (2026) demonstrate that RLVR restores pruned reasoning models to near-original accuracy, substantially outperforming SFT. We show that this hierarchy is an artifact of suboptimal pruning. When pruning preserves representational diversity, recovery method choice becomes irrelevant: at 20% sparsity, mean $|\Delta(\text{GRPO} - \text{SFT})|$=0.54 pp across the four fine-grained criteria (max 0.61 pp); including dose-response variations, MBPP, and MATH-500, the maximum is 1.8 pp. What matters is not *how* you train after pruning, but *which structures* you remove.

This gap is surprising because RL and SFT impose fundamentally different demands on model representations: RL compresses incorrect paths while SFT expands correct ones (Matsutani et al., 2026). Yet existing structured pruning methods for reasoning LLMs (Wang et al., 2026, 2025; Ma et al., 2023; Kolawole et al., 2025) optimize for post-pruning quality without studying how the criterion shapes downstream RL recovery. No prior work has compared multiple pruning criteria under RL or asked whether the "RLVR > SFT" advantage persists across pruning strategies. We note that Ji et al. (2026) evaluate unstructured magnitude pruning retaining 96.5% of dense parameters; our findings concern fine-grained structural pruning where entire channel groups are removed, and the two settings may impose different demands on RL recovery.

We present a controlled comparison on DeepSeek-R1-Distill-Qwen-7B (DeepSeek-AI, 2025) at 20% MLP channel sparsity, varying the balance between gradient importance and CKA-based representational diversity (Kornblith et al., 2019) via a coefficient α in the pruning score (Figure 1). The dose-response curve reveals an inverted-U: recovery peaks at α=2.0 (66.11% GRPO on GSM8K (Cobbe et al., 2021)), where importance and diversity reach optimal balance. Under GRPO (Shao et al., 2024) training, four

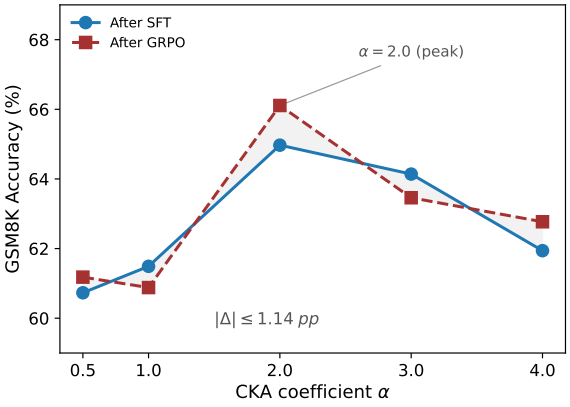


Figure 1: Dose-response curve for the CKA diversity coefficient α at 20% MLP channel sparsity on GSM8K (N=1,319). Recovery accuracy follows an inverted-U, peaking at α=2.0 (66.11%). SFT and GRPO curves nearly overlap ($|\Delta| \leq 1.14$ pp at every point), visually demonstrating reward-accuracy decoupling at this sparsity level: the pruning criterion, not the recovery paradigm, shapes outcome.

criteria produce a **recovery capacity spectrum**, ranging from active learning (CKA-only, peak reward 4.4× Standard) through stagnation (Random) to collapse (α=1.0), that tracks preserved representational diversity, not gradient importance. A depth pruning baseline (ShortGPT; Men et al., 2024) is the only method with $|\Delta| > 1$ pp on GSM8K (-1.36 pp), confirming that pruning granularity interacts with RL trainability.

Our contributions are:

1. **Reward-accuracy decoupling.** At 20–25% sparsity, GRPO and SFT produce near-identical accuracy across all four fine-grained criteria: mean $|\Delta|$=0.54 pp at 20% (max 0.61 pp; 1.8 pp including dose-response variations, MBPP, and MATH-500). At 15%, GRPO can still amplify the CKA-guided advantage (+1.51 pp differential), but this partial exception reinforces the main pattern: the pruning criterion, not the recovery paradigm, shapes outcome (§4.3).
2. **Recovery capacity spectrum.** We propose RASP, which balances gradient importance with CKA-based representational diversity via a coefficient α. A dose-response analysis on GSM8K reveals an inverted-U peaking at α=2.0 (66.11%). Under GRPO training, four criteria produce qualitatively distinct dynamics, ranging from active learning to full collapse, that track preserved diversity, not gradient importance. The spectrum has preliminary evidence at 1.5B scale and, via SFT-based recovery, across Qwen2.5-family models (Qwen2.5-Math-7B) (§3.3, §4.2, §4.5).

2 Related Work

Structured pruning of LLMs. Width pruning methods (LLM-Pruner (Ma et al., 2023), Bonsai (Kolawole et al., 2025), GRASPrune (Wang et al., 2026)) remove attention heads or FFN channels while preserving dense computation. Depth pruning (Men et al., 2024; Ashkboos et al., 2024; Yang et al., 2024b) removes entire layers but causes severe reasoning degradation (Shrestha et al., 2026); Monjur et al. (2026) revisit the trade-off under test-time scaling. For reasoning models, RESP (Wang et al., 2025), RAC (Lucas et al., 2025), and SSGR (Xiang et al., 2025) refine calibration data but retain gradient-based importance as the scoring criterion; Ding et al. (2026) confirm width pruning is more robust than depth pruning for reasoning but do not study RLVR recovery. A separate line uses functional structure to guide pruning: FANG (Yu et al., 2025) clusters FFN neurons by activation, FuncNet (Liu et al., 2025a) decomposes LLMs via ICA, and MPruner (Hu et al., 2024) applies inter-layer CKA (Kornblith et al., 2019) for depth pruning in CNNs. We differ in both granularity and objective: *intra-layer* CKA between individual structures, optimizing for RL recovery capacity rather than immediate accuracy.

Pruning-aware recovery. Ji et al. (2026) show that RLVR restores pruned reasoning models to near-original accuracy, outperforming SFT, but evaluate only importance-based unstructured pruning at low sparsity (96.5% retention). Under fine-grained structural pruning, our controlled experiments show that this advantage vanishes when pruning preserves diversity: it is a property of the pruning criterion, not of RL itself. Other work optimizes complementary axes: post-pruning scaling (P^2 Law; Chen et al., 2025), recovery data selection (PASER; He et al., 2025), joint pruning-tuning (PAT; Liu et al., 2025b), post-hoc attention compensation (RestoreLCC; Feng et al., 2025), iterative prune-tune loops (Zhao et al., 2026), post-RLVR compression (OPSD; Kim and Lee, 2026), and rollout pruning during RLVR (Xu et al., 2026)—but none study which *pruning criterion* enables RL-based recovery. The Lottery Ticket Hypothesis (Frankle and Carbin, 2019) asks which subnetworks train effectively from scratch; our recovery capacity spectrum poses the analogous question for

post-pruning recovery, where the answer depends on the recovery paradigm (RLVR vs. SFT).

RLVR training dynamics. Matsutani et al. (2026) show that RLVR *compresses* incorrect reasoning trajectories while SFT *expands* correct ones, predicting RL–SFT complementarity. Our results qualify this prediction: the asymmetry manifests as a recovery capacity difference (the spectrum), but the expected accuracy gap disappears when pruning preserves representational diversity ($|\Delta| \leq 1.8$ pp across GSM8K, MBPP, and MATH-500; §4.3). RL instability is well studied in the RLHF context (Gao et al., 2023; Dong et al., 2025; Shah et al., 2025); Mukherjee et al. (2025) find that RL fine-tuning updates only 5–30% of parameters, raising the question of whether pruning criteria interact with this active subnetwork. We connect these observations to structural pruning: the pruning criterion shapes susceptibility to RL collapse, but the resulting accuracy is invariant to the recovery paradigm at 20–25% sparsity.

3 Method

To isolate how the pruning criterion affects post-RLVR recovery, we hold everything else constant: a shared structural analysis computes importance and functional overlap scores for all prunable structures, four scoring criteria combine these scores differently to select structures for removal, and an identical two-stage recovery pipeline is applied to every pruned model. The only variable across conditions is which structures are removed (Figure 2).

3.1 Preliminaries and Notation

Structured pruning. Given a pretrained transformer $\mathcal{M}$ with L layers, structured pruning removes entire computational units (attention heads or MLP channels) to reduce model size while maintaining hardware-friendly dense computation. We denote the set of prunable structures as $\mathcal{H} = \{h_1, h_2, \ldots, h_N\}$.

Grouped Query Attention (GQA). In GQA architectures (Ainslie et al., 2023), n_Q query heads share n_{KV} key-value heads ($G = n_Q/n_{KV}$ per group). For DeepSeek-R1-Distill-Qwen-7B (n_Q=28, n_{KV}=4, G=7), we treat each GQA group (7 Q heads + 1 KV head) as a single prunable unit.

RLVR and GRPO. Reinforcement Learning with Verifiable Rewards (RLVR) recovers pruned model capabilities using outcome-based rewards from tasks with verifiable answers. We use Group Relative Policy Optimization (GRPO; Shao et al., 2024) as the recovery algorithm.

3.2 Structural Analysis

All four pruning criteria share a common structural analysis stage that computes two scores per structure: *importance* $I(h)$ capturing the structure's contribution to reasoning, and *functional overlap* $F(h)$ measuring representational redundancy with other structures in the same layer.

Calibration data. We compute importance and overlap scores on chain-of-thought reasoning data from GSM8K (Cobbe et al., 2021) training examples. Given a calibration set $\mathcal{D}_{\text{cal}} = \{(x_j, y_j)\}_{j=1}^{M}$ with M=512 randomly sampled examples, we compute activations and gradients on the concatenated sequences (x_j, y_j).

Structural importance. For each structure h with activation $\mathbf{a}_h$, we approximate the loss change from removal via first-order Taylor expansion ($\Delta\mathcal{L} \approx -\mathbf{a}_h^\top \nabla_{\mathbf{a}_h}\mathcal{L}$, dropping $O(\|\mathbf{a}_h\|^2)$ terms) and aggregate over the calibration set:

$$I(h) = \left| \sum_{(x,y)\in\mathcal{D}_{\text{cal}}} \mathbf{a}_h(x,y)^\top \frac{\partial \mathcal{L}(x,y)}{\partial \mathbf{a}_h} \right|, \qquad (1)$$

where $\mathcal{L}$ is the language modeling loss over the reasoning trace. For a GQA group g containing Q heads $\{q_1, \ldots, q_G\}$ and KV head k, we aggregate: $I(g) = \sum_{i=1}^{G} I(q_i) + I(k)$.

Functional overlap via CKA. We measure functional overlap using linear Centered Kernel Alignment (CKA; Kornblith et al., 2019), which quantifies similarity between two sets of representations invariant to invertible linear transformations. Given activation matrices $\mathbf{X}, \mathbf{Y} \in \mathbb{R}^{n\times d}$ collected over n calibration examples, linear CKA is:

$$\text{CKA}(\mathbf{X}, \mathbf{Y}) = \frac{\text{HSIC}(\mathbf{X}, \mathbf{Y})}{\sqrt{\text{HSIC}(\mathbf{X}, \mathbf{X}) \cdot \text{HSIC}(\mathbf{Y}, \mathbf{Y})}}, \qquad (2)$$

where $\text{HSIC}(\mathbf{X}, \mathbf{Y}) = \frac{1}{(n-1)^2}\,\text{tr}(\mathbf{KHLH})$, with $\mathbf{K} = \mathbf{X}\mathbf{X}^\top$, $\mathbf{L} = \mathbf{Y}\mathbf{Y}^\top$, and $\mathbf{H} = \mathbf{I}_n - \frac{1}{n}\mathbf{1}\mathbf{1}^\top$.

For a GQA group g_i in layer ℓ, we form its representation by concatenating its constituent Q head activations: $\mathbf{X}_{g_i} = [\mathbf{a}_{q_1}; \ldots; \mathbf{a}_{q_G}] \in \mathbb{R}^{n\times Gd_h}$. The functional overlap of g_i is the mean pairwise CKA

with other groups in the same layer:

$$F(g_i) = \frac{1}{|\mathcal{G}_\ell| - 1} \sum_{\substack{g_j \in \mathcal{G}_\ell \\ j \neq i}} \text{CKA}(\mathbf{X}_{g_i}, \mathbf{X}_{g_j}). \quad (3)$$

We apply the same intra-layer CKA overlap to MLP channel groups, partitioning each MLP intermediate dimension into equal-sized groups. Unlike prior work that uses inter-layer CKA to identify globally redundant layers for depth pruning (Hu et al., 2024; Lachance, 2022), we use *intra-layer* CKA between individual structures to measure per-structure functional redundancy. Computing all pairwise CKA scores adds negligible overhead ($O(L \cdot k^2)$, $k \leq 28$; <2 min on one GPU).

CKA reliability. Our intra-layer use of linear CKA between structures of identical dimensionality avoids the kernel width and layer width sensitivities identified by Davari et al. (2023); Klabunde et al. (2023); empirically, structure rankings are stable across calibration sizes (Spearman $\rho \geq 0.964$; Appendix H).

3.3 Pruning Scoring Criteria

We compare four scoring criteria that combine importance $I(h)$ and functional overlap $F(h)$ in different ways. All four methods use the same per-layer capacity floor (≥40% of MLP groups per layer must be retained, chosen to prevent any single layer from becoming a throughput bottleneck; ablation over $\tau \in \{0.3, 0.4, 0.5, 0.6\}$ in Appendix S) and global ranking procedure: structures are sorted by score, and the lowest-scoring structures are pruned up to the target sparsity ratio, skipping any candidate whose removal would violate the layer floor.

Standard (importance-only). Following LLM-Pruner (Ma et al., 2023), structures are ranked solely by importance:

$$S_{\text{std}}(h) = I(h). \quad (4)$$

This is the industry-standard criterion that retains gradient-important structures and removes "unimportant" ones.

RASP (recovery-aware composite). The recovery-aware score combines importance and functional overlap:

$$S_{\text{rasp}}(h) = I(h) \times \big(1 - F(h)\big)^{\alpha}, \quad (5)$$

where $\alpha \geq 0$ controls the strength of the overlap penalty (α=0 reduces to Standard). Structures with high importance *and* low overlap receive the highest scores and are retained; structures with high overlap are penalized and pruned preferentially, regardless of their importance.

CKA-only (overlap-only). CKA-only scoring ignores gradient importance entirely, ranking structures by functional overlap alone. Structures with the highest mean pairwise CKA similarity to other structures in their layer are removed first:

$$S_{\text{cka}}(h) = 1 - F(h). \quad (6)$$

This criterion preserves all representationally unique structures regardless of their gradient importance, removing only those whose function is redundant with retained structures. S_{cka} corresponds to the limit $\alpha \to \infty$ in Eq. 5.

Random. Structures are selected uniformly at random for removal, subject to the per-layer capacity floor. This baseline controls for the effect of any informed scoring criterion.

The four criteria span a spectrum from pure gradient importance (Standard) to pure representational diversity preservation (CKA-only), with RASP as a weighted interpolation and Random as an uninformed control. The key distinction is that CKA-only protects *functionally unique* structures regardless of gradient contribution: structures deemed "unimportant" by gradient metrics may provide non-redundant computational pathways essential for RL exploration (Matsutani et al., 2026).

3.4 Two-Stage Recovery

After pruning, all four models undergo the same two-stage recovery pipeline with identical GRPO configuration, except that Random uses the TRL default generation limit (512 vs. 1,024 tokens; Section 4.1), making it a conservative comparison point.

Stage 1: SFT warmup. Structured pruning disrupts the model's ability to generate end-of-sequence (EOS) tokens: in sampling mode, pruned models produce repetitive outputs without termination. Since GRPO computes rewards only from completed generations, this EOS collapse makes direct RLVR recovery impossible. We restore generation competence through a short SFT phase on chain-of-thought reasoning demonstrations with teacher forcing. This stage is intentionally brief:

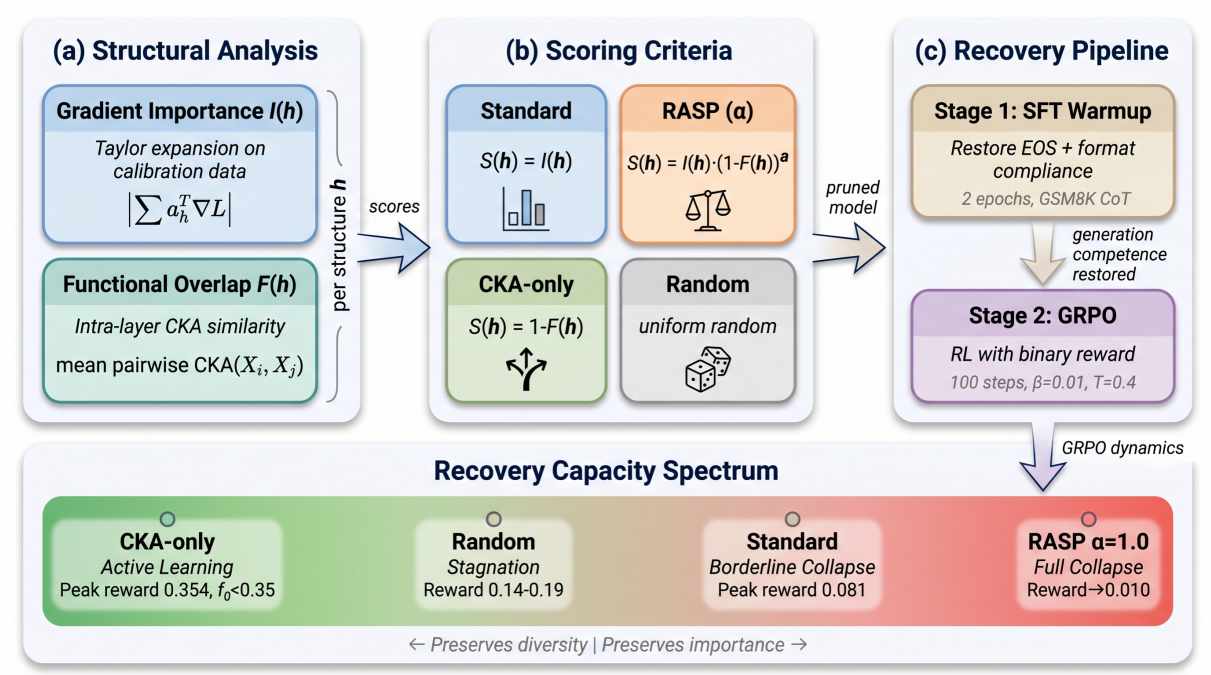


Figure 2: Overview of our pruning comparison framework. **Left**: structural analysis computes importance $I(h)$ and CKA-based overlap $F(h)$ scores, which four scoring criteria combine differently (Eqs. 4–6). **Center**: ranked structures are removed up to the target sparsity, subject to per-layer capacity floors. **Right**: all pruned models undergo identical two-stage recovery (SFT warmup → GRPO). **Bottom**: the resulting recovery capacity spectrum, from active learning (CKA-only) to full collapse (RASP α=1.0), tracks preserved representational diversity rather than gradient importance.

its purpose is to restore the minimal capabilities (EOS generation, format compliance) needed for GRPO, not to recover full reasoning ability.

Stage 2: GRPO alignment. With generation competence restored, we apply GRPO on mathematical reasoning problems with binary reward (r=1 if the final answer matches the ground truth, r=0 otherwise). RLVR can globally explore and redistribute functionality across remaining structures (Matsutani et al., 2026), while SFT is limited to imitating fixed trajectories through a reduced architecture. The GRPO configuration (learning rate, KL coefficient, sampling temperature, loss formulation) is identical across all four pruned models, except that Random uses a shorter generation limit (Section 4.1).

3.5 Recovery Capacity Diagnostics

We define **recovery capacity** as the model's ability to benefit from post-pruning recovery training, operationalized as peak mean reward over the GRPO training horizon. Four diagnostics (mean reward, zero-variance batch fraction (f_0), gradient norm, and KL divergence) distinguish three regimes: *active learning* (rising reward, low f_0), *stagnation* (flat reward), and *collapse* (declining reward, high f_0, vanishing gradients).

The complete pruning comparison procedure is given in Algorithm 1 (Appendix B).

4 Experiments

4.1 Experimental Setup

Base model and evaluation. We evaluate on DeepSeek-R1-Distill-Qwen-7B (DeepSeek-AI, 2025), a 7B-parameter reasoning model built on the Qwen2.5 architecture (Yang et al., 2024a) with GQA (Ainslie et al., 2023) (architecture details in Section 3.1). Our primary benchmark is GSM8K (Cobbe et al., 2021), a set of 1,319 grade-school math problems requiring multi-step arithmetic reasoning. We report exact-match accuracy: the final numerical answer is extracted from the model's generation and compared against the ground truth. For rapid iteration during development, we use a 100-problem subset (N=100); final results use the full test set (N=1,319). The original unpruned model achieves 90.83% on GSM8K (greedy decoding).

Pruning methods. We compare four pruning criteria at 20% structured sparsity, all removing the same number of MLP channel groups (200 of 784 across 28 layers, ∼1.45B parameters) with identical per-layer capacity floors (≥40% retention per layer):

1. **CKA-only**: structures ranked by functional overlap $F(h)$ alone (Eq. 3); the most redundant groups are removed regardless of gradient importance.
2. **Random**: uniform random selection among eligible groups.
3. **Standard**: Taylor-expansion importance $I(h)$ alone (Eq. 1); the least important groups are removed, following LLM-Pruner (Ma et al., 2023).
4. **RASP** (α=1.0): the composite score $S(h) = I(h) \times (1 - F(h))^{\alpha}$ (Eq. 5), removing structures that are both low-importance and functionally redundant.

Recovery pipeline and hardware. All four methods undergo matched two-stage recovery. Stage 1 (SFT warmup): 2 epochs on 7,473 GSM8K chain-of-thought examples with teacher forcing (37 minutes), restoring EOS generation and basic format compliance. Stage 2 (GRPO): Group Relative Policy Optimization (Shao et al., 2024) with learning rate 1×10^{-6}, maximum completion length 1,024 tokens, KL penalty coefficient β=0.01, sampling temperature 0.4, DAPO-style clipped loss, and binary reward (r=1 if correct, r=0 otherwise), trained for 100 steps on 1,000 GSM8K training problems. All methods use identical GRPO configuration with `max_completion_length`=1,024, except Random which uses the TRL default of 512 tokens (Table 3; Appendix A), making it a conservative comparison point. Seeds vary at the GRPO stage; pruning calibration and SFT use identical configurations across all methods. We repeat GRPO training with 3 random seeds for CKA-only, Standard, and Random (Table 2). CKA-only achieves 64.39% ± 0.19 pp (std across seeds), Standard 59.84% ± 0.38 pp; the 4.55 pp gap is robust across seeds (Welch's t-test, p=0.0009, Cohen's d=11.1; 95% CI for mean difference: [3.62, 5.47] pp), confirming that recovery capacity differences are structural rather than stochastic (per-seed results in Appendix J). All experiments run on 2× NVIDIA RTX PRO 6000 GPUs (94 GB VRAM each); pruning variants are trained in parallel on separate GPUs under identical wall-clock conditions.

Table 1: Recovery capacity spectrum: GRPO dynamics for four pruning criteria at 20% sparsity under identical configuration (lr 1×10^{-6}, β=0.01, T=0.4, 1,024-token limit, 100 steps); Random uses TRL default 512-token limit, making it a conservative comparison point. f_0: fraction of batches where all completions receive identical reward (lower = healthier). Best in **bold**.

Method	Peak Rew. ↑	Mean Rew. ↑	Min f_0 ↓	Behavior
CKA-only	**0.354**	**0.285**	**0.125**	Active learning
Random	0.190	0.164	0.300	Stagnation
Standard	0.081	0.050	0.500	Borderline collapse
RASP (α=1.0)‡	0.061	0.031	0.675	Full collapse

‡Single seed (N=1). CKA-only, Standard, and Random validated with 3 seeds (Table 2).

Table 2: GRPO responsiveness on GSM8K (N=1,319, greedy decoding), evaluated at step 100. All methods share identical GRPO configuration (Table 3), except Random which uses TRL default 512-token limit. CKA-only, Standard, and Random report mean ± std over 3 GRPO seeds; RASP uses a single seed. Best in **bold**.

Model	SFT Acc. (%)	GRPO Acc. (%)	Δ (pp)	Peak Rew.
Unpruned‡	**81.27**	**81.27**	0.00	**0.669**
CKA-only 20%	63.91	64.39±0.19	+0.48	0.354
Standard 20%	59.29	59.84±0.38	+0.55	0.081
RASP (α=1.0) 20%	61.49	60.88	−0.61	0.061
Random 20%	51.25	50.72±0.15	−0.53	0.190
ShortGPT 20%†	44.73	43.37	−1.36	–

†Depth pruning baseline (Men et al., 2024); removes entire layers.
‡All rows including the dense baseline use SFT-warmup checkpoints; the pre-warmup dense model achieves 90.83% on GSM8K.

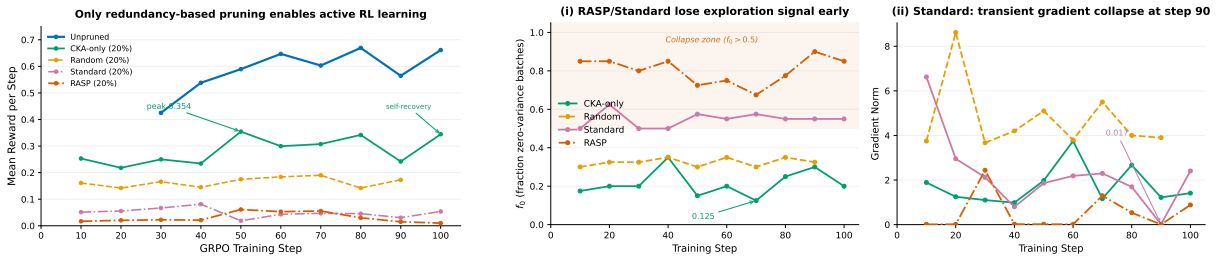
4.2 Recovery Capacity Spectrum

Under GRPO training, the four pruning criteria produce a clear recovery capacity spectrum: CKA-only ≫ Random ≫ Standard > RASP α=1.0 (Table 1, Figure 3). The pruning criterion, not training configuration, shapes whether the model learns, stagnates, or collapses.

Among these configurations, only the diversity-maximizing criterion (CKA-only) produces active learning: peak reward 0.354 with f_0 as low as 0.125 (Figure 3b). Random stagnates (reward 0.14–0.19, min $f_0 = 0.300$); Standard exhibits borderline collapse ($f_0 \geq 0.50$, peak reward 0.081); RASP at α=1.0 collapses ($f_0 \geq 0.675$, reward declining to 0.010, with near-zero KL divergence indicating minimal policy movement; Appendix Q). The dose-response (§4.4) confirms CKA weight as the critical variable: increasing α from 1.0 to 2.0 shifts RASP from collapse to peak recovery (66.11% vs. 64.39% for CKA-only on GSM8K).

4.3 Reward-Accuracy Decoupling

Despite the 4.4× peak reward gap, GRPO barely moves accuracy (Figure 4): all fine-grained meth-



(a) Reward trajectories

(b) f_0 and gradient norm diagnostics

Figure 3: GRPO training dynamics for four pruning criteria (100 steps, β=0.01, T=0.4). (a) Reward: CKA-only reaches 0.354 peak; Random oscillates (0.14–0.19); Standard peaks at 0.081; RASP collapses to 0.010. (b) Left: f_0 (zero-variance batch fraction); CKA-only stays below 0.35 while Standard and RASP remain in the collapse zone ($f_0 > 0.5$). Right: gradient norm; RASP norms near zero throughout; Standard shows a transient minimum at step 90.

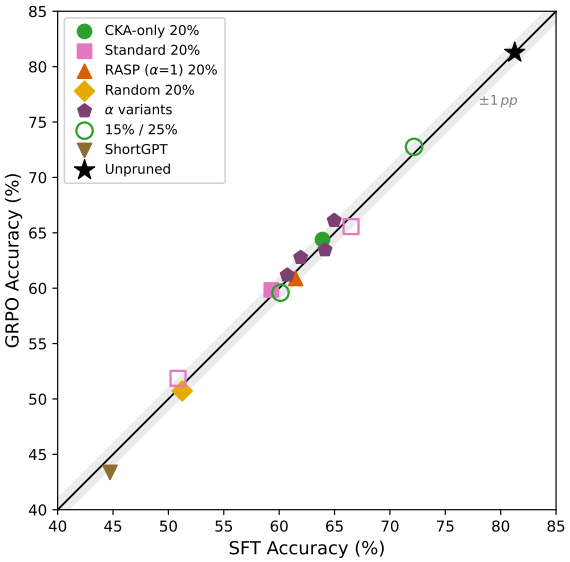


Figure 4: SFT vs GRPO accuracy across all pruning criteria and sparsity levels on GSM8K (N=1,319). Points near the y=x line indicate reward-accuracy decoupling: despite 4.4× peak reward gaps (Figure 3a), GRPO barely moves accuracy ($|\Delta| \leq 1$ pp for all fine-grained methods at 20–25%).

ods at 20% sparsity change by $|\Delta| < 1$ pp (CKA-only +0.48, Standard +0.55, Random −0.53, RASP −0.61 pp; Table 2). We term this *reward-accuracy decoupling*: reward trajectory reflects the model's capacity to learn from RL signal, not downstream accuracy. At 20% sparsity, the four fine-grained criteria show mean $|\Delta|$=0.54 pp (max 0.61 pp); across the full α dose-response spectrum, $|\Delta|$ remains below 1.14 pp.

The degree of decoupling varies with sparsity: at 15%, GRPO slightly amplifies the CKA-only advantage (SFT gap 5.69 pp → GRPO gap 7.20 pp); at 20%, GRPO adds nothing (gap preserved at 4.55 pp); at 25%, Standard's peak reward collapses to 0.008 yet accuracy still changes by only +0.99 pp. Across all four fine-grained criteria at 20% and 25%, $|\Delta| < 1$ pp without exception on GSM8K (Tables 2, 12; Appendix M); dose-response α variations show slightly larger but still small differences (max 1.14 pp at α=2.0; Table 5; MATH-500 differences within statistical noise, Appendix F).

TOST equivalence testing (ε=2 pp, N=1,319 paired examples) confirms statistical equivalence for 7/8 comparisons (p<0.05; Appendix T), robust across seeds (0.37 pp range; Appendix L) and extended training (300 steps; §4.5). The endpoints of the spectrum generalize beyond GSM8K: on MATH-500 (Hendrycks et al., 2021), CKA-only and Standard both rank above RASP and Random in SFT accuracy (27.4%, 25.6% vs. 21.6%, 16.6%), though the CKA-only–Standard gap narrows (1.8 pp vs. 4.62 pp on GSM8K; Appendix F), possibly because MATH-500's higher difficulty compresses all pruned methods toward a floor. On MBPP (Austin et al., 2021) all fine-grained GRPO $|\Delta| < 2$ pp, though not statistically significant at this sample size (Appendix G). ShortGPT (Men et al., 2024) depth pruning is the sole exception: its GRPO delta is task-dependent (−1.36 pp on GSM8K, +4.01 pp on MBPP; Tables 2, 7), unlike fine-grained methods ($|\Delta| \leq 1.8$ pp).

4.4 Additional Results

Post-SFT accuracy does not predict GRPO outcome. Under SFT alone, CKA-only (63.91%) > RASP (61.49%) > Standard (59.29%) > Random (51.25%), yet under GRPO the RASP/Standard ordering reverses and Random outperforms both in peak reward. The spectrum is driven by structural

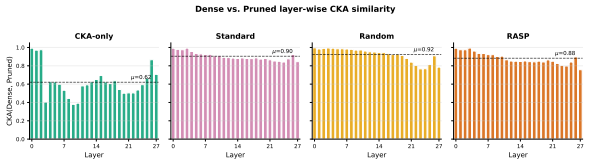


Figure 5: Layer-wise CKA similarity between dense and pruned representations (28 layers). CKA-only diverges most from dense (μ=0.62) yet recovers best; Standard (μ=0.90), Random (μ=0.92), and RASP (μ=0.88) preserve dense fidelity but collapse under RL.

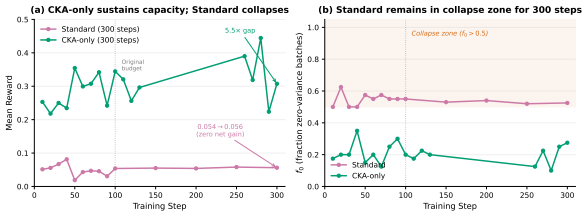


Figure 6: Extended GRPO training (300 steps, 3× budget). (a) Standard reward: zero net gain (0.054→0.056); CKA-only sustains a 5.5× gap. (b) Standard f_0 remains in the collapse zone (>0.5); CKA-only stays below 0.30.

properties of the pruned architecture, not post-SFT accuracy.

Dose-response and sparsity limits. The CKA coefficient α continuously controls recovery capacity. On GSM8K, recovery follows an inverted-U across $\alpha \in \{0.5, 1.0, 2.0, 3.0, 4.0\}$, peaking at α=2.0 (SFT 64.97%, GRPO 66.11%; Figure 1, Table 5) with $|\Delta| \leq 1.14$ pp at every point; α=1.0 collapses (peak reward 0.061) while α=4.0 recovers to 62.77% (Appendix D).[1] The key finding generalizes to MATH-500: α=1.0 remains the worst (GRPO 23.0%), while higher-α and CKA-only methods cluster near the top (GRPO 27.2–27.6%), though the inverted-U shape is not resolved at this difficulty level (Table 6; Appendix F). All methods collapse at 30% sparsity (Appendix K); pruning reduces parameters by 19% and memory by ∼28% (Appendix C).

4.5 Extended Analysis

Representational diversity. CKA similarity between dense and pruned representations (Figure 5) shows CKA-only diverges most from the dense model (μ=0.62), while Standard (μ=0.90), Random (μ=0.92), and RASP (μ=0.88) cluster near 1.0. Preserving dense fidelity through gradient importance appears to retain redundant representations at the expense of the diversity that GRPO exploration requires. CKA-only concentrates pruning in early layers (L0–9, 72.5% of removed groups) while Standard prunes uniformly, removing late-layer structures that are representationally unique (Appendix P, R). Gradient flow analysis corroborates this: CKA-only models exhibit 2.89× mean gradient magnitude versus Standard across 55/56 layer groups (p<0.05, Wilcoxon; Appendix I), consistent with diversity preservation enabling stronger RL training signals. On MATH-500 (Table 6), CKA-only truncates only 22% of completions (mean 655 tokens) versus Standard's 44% (mean 1,128 tokens), yet achieves equal or higher accuracy; diversity preserves efficient reasoning rather than redundant chains.

Extended training cannot rescue collapse. Extending GRPO from 100 to 300 steps (Figure 6; Appendix E), Standard shows zero net improvement (reward 0.054–0.058, $f_0 > 0.5$ throughout), while CKA-only sustains a 5.5× gap (0.308 vs 0.056 at step 300). The gap is structural, not a matter of training budget.

Cross-scale and cross-model generalization. Preliminary evidence at 1.5B scale (Table 13; Appendix N) preserves the three-method ordering. On Qwen2.5-Math-7B, a different model within the Qwen2.5 family (Appendix O), CKA-only recovers from 1.0% post-prune to 51.50% vs. Standard's 21.00%.

5 Discussion and Conclusion

Whether RLVR outperforms SFT for recovering pruned reasoning models depends on how the model was pruned. Under diversity-preserving pruning at 20–25% sparsity, the two paradigms produce near-identical accuracy (mean $|\Delta|$=0.54 pp; max 1.8 pp), yet the pruning criterion creates a 4.4× peak reward gap. SFT warmup performs the bulk of accuracy recovery (∼60 pp); GRPO exposes structural capacity differences but adds negligible accuracy, a pattern confirmed at 300 steps where CKA-only reward rises yet accuracy remains unchanged. CKA-only preserves distinct pathways that sustain RL exploration (§4.5); the non-monotonic α dose-response argues against a layer-distribution confound.

[1] α=1.0 uses `max_completion_length`=1,024; other α values use the TRL default of 512. Since α=1.0 is the curve minimum, this makes the inverted-U a conservative estimate.

Limitations

Architecture and scale. Our primary results are on DeepSeek-R1-Distill-Qwen-7B (DeepSeek-AI, 2025) with GQA (G=7). Cross-model validation on Qwen2.5-Math-7B covers SFT recovery only (N=200); both models share the Qwen2.5 base architecture, so our cross-model evidence does not extend to different architecture families (e.g., LLaMA, Mistral). Whether the spectrum generalizes across architecture families and scales (1B–70B) remains open.

Diagnostic limitations. Reward-accuracy decoupling at 20–25% sparsity means that recovery capacity, operationalized as peak reward, does not predict downstream accuracy improvements within our training horizon. Recovery capacity captures the model's ability to sustain RL training without collapse, but practitioners seeking accuracy gains may require longer training, different reward formulations, or lower sparsity levels. The per-layer capacity floor (40% retention) was chosen heuristically; an ablation over $\tau \in \{0.3, 0.4, 0.5, 0.6\}$ (Appendix S) suggests moderate sensitivity, but a full search over continuous τ values was not performed.

Inference efficiency. At 20% sparsity, pruning reduces memory by ∼28% and improves time-to-first-token, but inference throughput decreases by 19% (72.0→58.3 tok/s), likely because channel-group pruning creates irregular sparse patterns that current hardware cannot accelerate.

Replication and scope. CKA-only, Standard, and Random are validated over 3 GRPO seeds; RASP uses a single seed (N=1). We evaluate on three benchmarks (GSM8K, MATH-500, MBPP) covering math reasoning and code generation; logical, commonsense, and multi-modal reasoning are not addressed.

Ethics Statement

This work uses publicly available models (DeepSeek-R1-Distill-Qwen, Qwen2.5-Math) and established benchmarks (GSM8K, MATH-500, MBPP); no private data or human subjects are involved. Structured pruning reduces model size and inference memory, which can lower the computational cost and energy consumption of deploying large language models. Our research focuses on the technical analysis of pruning criteria and their interaction with reinforcement learning recovery; we do not foresee direct negative societal impacts from this work. All models and datasets used in this work are publicly available under permissive licenses: DeepSeek-R1-Distill-Qwen (MIT), Qwen2.5-Math (Apache 2.0), GSM8K (MIT), MATH (MIT), and MBPP (CC-BY-4.0).

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A Training Details

B Algorithm

C Inference Efficiency

All pruned models achieve ∼28% memory reduction from structural parameter removal. TTFT (time to first token) improves by 18% (20.1→16.4 ms), reflecting faster prefill with fewer parameters. Per-token throughput decreases by 19% (72.0→58.3 tok/s, or 13.9→16.7 ms per token), consistent with non-uniform layer widths creating computational bottlenecks that current GPU kernels do not optimize away. CKA-only and Standard show identical throughput (58.3 vs 58.2 tok/s), confirming that the pruning criterion affects recovery capacity but not inference speed at matched sparsity.

Table 3: Hyperparameter details for SFT warmup and GRPO recovery stages. All methods share the same GRPO configuration; Random uses the TRL default generation limit (marked with †).

Parameter	SFT	GRPO
Learning rate	2×10^{-5}	1×10^{-6}
Batch size	1	1
Gradient accumulation	8	1
Max sequence length	2,048	–
Max new tokens	–	1,024 †512
Optimizer	AdamW	AdamW
Epochs / Steps	2 epochs	100 steps
KL coefficient β	–	0.01
Sampling temperature	–	0.4
Num generations per prompt	–	8
Reward	–	Binary (exact match)
Loss formulation	Cross-entropy	DAPO-style clipped
`reward_desert_patience`	–	50

†Random uses TRL default of 512 tokens; all other methods use 1,024.

Table 4: Inference characteristics at 20% sparsity (single-GPU, no contention). Pruning reduces parameters by 19% and memory by ∼28%, with TTFT improving 18%. Per-token throughput decreases 19% (0.81×) due to non-uniform layer widths from full-layer removal that current GPU kernels do not optimize.

Configuration	TPS (tok/s)	TTFT (ms)	Per-token (ms)	Params (B)
Dense 7B	72.0	20.1	13.9	7.62
CKA-only 20%	58.3	16.4	16.7	6.16
Standard 20%	58.2	16.4	16.7	6.16

D CKA Coefficient α Ablation

On GSM8K, the dose-response curve across $\alpha \in \{0.5, 1.0, 2.0, 3.0, 4.0\}$ reveals an inverted-U pattern peaking at α=2.0 (SFT 64.97%, GRPO 66.11%, Δ=+1.14 pp; Table 5). At α=0.5, importance still dominates; recovery is low (SFT 60.73%, GRPO 61.18%). At α=1.0 (RASP), GRPO collapses (peak reward 0.061, accuracy 60.88%, Δ=−0.61 pp). α=3.0 (SFT 64.14%, GRPO 63.46%, Δ=−0.68 pp) excludes a plateau hypothesis: accuracy drops 2.65 pp from the α=2.0 peak, confirming that the optimum is a true peak rather than a saturation regime. At α=4.0, CKA overlap dominates; the model recovers to 62.77% GRPO accuracy (SFT 61.94%, Δ=+0.83 pp) with active GRPO learning (reward 0.146→0.256→0.198). Reward-accuracy decoupling holds at every α value: $|\Delta| \leq 1.14$ pp across all five points.

Algorithm 1 Controlled Pruning Comparison Framework

Require: Model $\mathcal{M}$, calibration data $\mathcal{D}_{\text{cal}}$, target sparsity s, scoring criterion $\mathcal{C} \in \{\text{Std}, \text{RASP}, \text{CKA}, \text{Rand}\}$, layer floor τ
Ensure: Pruned and recovered model $\mathcal{M}'$

% Stage 1: Shared Structural Analysis
for each structure $h \in \mathcal{H}$ **do**
  Compute importance $I(h)$ via Eq. (1)
  Collect activations $\mathbf{X}_h$ on $\mathcal{D}_{\text{cal}}$
for each layer $\ell = 1, \ldots, L$ **do**
  for each structure h_i in layer ℓ **do**
    $F(h_i) \leftarrow \frac{1}{|\mathcal{H}_\ell|-1} \sum_{j \neq i} \text{CKA}(\mathbf{X}_{h_i}, \mathbf{X}_{h_j})$
% Stage 2: Criterion-Specific Scoring
for each structure $h \in \mathcal{H}$ **do**
  $S(h) \leftarrow \mathcal{C}(I(h), F(h))$ ▷ Eq. 4, 5, 6, or random
Sort $\mathcal{H}$ by $S(h)$ ascending; prune lowest up to s, subject to τ
% Stage 3: Two-Stage Recovery
$\mathcal{M}_{\text{warm}} \leftarrow \text{SFT}(\mathcal{M}_{\text{pruned}}, \mathcal{D}_{\text{sft}})$ ▷ restore EOS & format
$\mathcal{M}' \leftarrow \text{GRPO}(\mathcal{M}_{\text{warm}}, \mathcal{D}_{\text{rlvr}})$ ▷ recover reasoning
return $\mathcal{M}'$

E Extended Training Dynamics

See Figure 6 in the main text for the 300-step extended training results.

F MATH-500 Out-of-Distribution Evaluation

CKA-only and Standard both outperform RASP and Random on MATH-500 (SFT: 27.4%, 25.6% vs. 21.6%, 16.6%; GRPO: 27.6%, 27.0% vs. 23.0%, 15.0%), consistent with GSM8K. However, the CKA-only–Standard gap narrows substantially: from 4.6 pp (GSM8K SFT) to 1.8 pp (MATH-500 SFT) to 0.6 pp (MATH-500 GRPO). Middle-ranked methods also swap positions (Standard > RASP on MATH-500 vs. RASP > Standard on GSM8K). This is consistent with MATH-500's higher difficulty compressing all pruned methods toward a capability floor, reducing inter-method differentiation (§4.3). CKA-only also produces more efficient reasoning: only 22% of completions are truncated at the 2,048-token limit (mean 655 tokens) versus Standard's 44% (mean 1,128 tokens), indicating that the preserved structure enables shorter reasoning chains rather than longer ones. The α dose-response effect is not statistically significant on MATH-500: SFT accuracy varies by $\sim$8 pp across α values, but with stderr $\approx$2 pp per point, attributable to high truncation rates at the 2048-token limit and floor effects at this difficulty level. The inverted-U pattern is validated on GSM8K (Table 5); MATH-500 does not resolve the dose-response shape.

Table 5: Dose-response analysis: effect of CKA coefficient α on GSM8K accuracy (N=1,319) at 20% MLP channel sparsity. α=0 reduces to Standard (importance-only); higher α weights representational diversity. Best in **bold**.

α	SFT Acc. (%)	GRPO Acc. (%)	Δ (pp)
0.5	60.73	61.18	+0.45
1.0	61.49	60.88	−0.61
2.0	**64.97**	**66.11**	+1.14
3.0	64.14	63.46	−0.68
4.0	61.94	62.77	+0.83

Table 6: MATH-500 out-of-distribution evaluation (N=500, greedy decoding, `max_new_tokens=2048`). CKA-only and Standard both outperform RASP and Random; the CKA-only–Standard GRPO gap is 0.6 pp (vs. 4.6 pp on GSM8K), consistent with a difficulty-driven floor effect. Best per section in **bold**.

Method	SFT (%)	GRPO (%)	Δ (pp)
Dense		53.0 †	
20% sparsity — method comparison			
CKA-only	**27.4**	**27.6**	+0.2
Standard	25.6	27.0	+1.4
RASP (α=1.0)	21.6	23.0	+1.4
Random	16.6	15.0	−1.6
20% sparsity — α dose-response			
α=0.5	26.8	26.8	+0.0
α=2.0	25.6	**27.4**	+1.8
α=3.0	23.6	–	–
α=4.0	**27.8**	27.2	−0.6
25% sparsity			
CKA-only	18.8	17.6	−1.2
Standard	15.4	–	–

†Over half of Dense responses truncated at 2048 tokens (73.6% at 4096); all pruned models evaluated under the same constraint.

G MBPP Code Generation

MBPP GRPO deltas are uniformly slightly positive across all fine-grained methods (mean +1.25 pp), but none are statistically significant at this sample size (N=374; stderr $\approx$1.7 pp). This directional consistency may reflect a weak task-specific GRPO effect at the capability floor rather than a violation

Table 7: MBPP code generation accuracy (N=374, greedy decoding). All fine-grained pruning methods converge near a capability floor (∼75% code capability loss vs Dense), limiting differentiation (∼1 pp range, within stderr). ShortGPT shows a larger GRPO delta (+4.01 pp) but at the 95% CI boundary (stderr ≈1.55 pp).

Method	SFT (%)	GRPO (%)	Δ (pp)
Dense	–	53.48	–
CKA-only 20%	12.83	14.17	+1.34
Standard 20%	12.03	13.10	+1.07
RASP (α=1.0) 20%	11.76	13.10	+1.34
ShortGPT 20%†	10.43	14.44	+4.01

†Depth pruning; removes entire layers.

of reward-accuracy decoupling.

H CKA Scoring Stability

Table 8: CKA scoring stability across calibration sizes. Spearman ρ between structure rankings produced by different calibration set sizes. High correlation ($\rho \geq 0.964$) indicates that CKA-based scoring is robust to calibration data volume.

Calibration Sizes	Spearman ρ (min)	Spearman ρ (mean)
256 / 512 / 1024	0.964	0.979

CKA-based structure rankings are highly stable across calibration data volumes: Spearman ρ between rankings from 256, 512, and 1024 calibration samples ranges from 0.964 to 0.979, indicating that the scoring criterion is not sensitive to calibration set size.

I Gradient Signal Analysis

We compare per-layer gradient magnitudes during early GRPO training (N=200 GSM8K calibration samples) across three pruning criteria using paired Wilcoxon signed-rank tests.

Table 9: Pairwise gradient magnitude ratios during GRPO training. Ratio: mean gradient magnitude of method A divided by method B across 56 parameter groups (28 layers × 2 projections). Significant layers: count where paired Wilcoxon signed-rank p<0.05.

Comparison	Gradient Ratio	Sig. Layers
CKA-only vs Standard	2.89×	55 / 56
CKA-only vs RASP	2.26×	56 / 56
RASP vs Standard	1.28×	45 / 56

The gradient magnitude hierarchy (CKA-only ≫ RASP > Standard) mirrors the recovery capacity hierarchy (GSM8K: 64.39% > 60.88% > 59.84%), with bootstrap 95% CIs for all ratios excluding 1.0. This is consistent with diversity preservation enabling stronger training signals during GRPO, though the evidence is correlational: an alternative explanation (that CKA-only retains more functional structure, producing higher initial loss and naturally larger gradients) cannot be ruled out.

Table 10: Sparsity cliff at 30% structured sparsity. All tested pruning criteria collapse, with near-zero accuracy and EOS generation rates below 6%.

Method	N	Acc. (%)	EOS Rate (%)
Standard	200	4	5.5
RASP (α=1.0)	200	0	3.0
CKA-only	500†	1.6	–†

†Terminated early after collapse confirmed; 8/500 correct.

J Seed Reproducibility

CKA-only GSM8K accuracy across 3 GRPO seeds: 64.37% (seed 42), 64.22% (seed 123), 64.59% (seed 456), yielding mean 64.39% ± 0.19 pp. Standard: mean 59.84% ± 0.38 pp across the same 3 seeds. See Table 11 for the full summary.

K Sparsity Cliff at 30%

All tested methods collapse at 30% sparsity: Standard retains 4% accuracy, CKA-only 1.6%, and RASP 0%. This capacity threshold, independent of pruning criterion, validates 20% as the operating point and confirms that the recovery capacity spectrum reflects how methods allocate a limited pruning budget.

L Multi-Seed GRPO Results

Table 11: Multi-seed GRPO results on GSM8K (N=1,319). CKA-only, Standard, and Random each trained with 3 GRPO seeds (42, 123, 456); pruning calibration and SFT are identical across all runs within each sparsity level.

Sparsity	Method	Mean Acc. (%)	Std (pp)	Seeds
20%	CKA-only	64.39	0.19	3
	Standard	59.84	0.38	3
	RASP	60.88	–	1
	Random	50.72	0.15	3
15%	CKA-only	72.76	0.57	3
	Standard	65.56	0.50	3

At 20% sparsity, the 4.55 pp gap between CKA-only and Standard is robust across seeds, with non-overlapping ranges (CKA-only: 64.22–64.59%, Standard: 59.44–60.20%). Standard shows a wider observed range across seeds (0.76 pp vs 0.37 pp), consistent with borderline-collapse dynamics where small perturbations produce larger outcome variation. Random (3 seeds: 50.57%, 50.72%, 50.87%; σ=0.15 pp, observed range 0.30 pp) converges in a narrow band with SFT accuracy seed-independent at 51.25%. All three seeds satisfy GRPO $\leq$ SFT (Δ=-0.68, -0.53, -0.38 pp), confirming that reward-accuracy decoupling is reproducible across seeds even for stagnating methods. Note: Standard seed 42 was trained from a separately initialized SFT checkpoint (accuracy 59.21% vs. 59.29% for seeds 123/456; $\Delta = 0.08$ pp). At 15% sparsity, the gap widens to 7.20 pp (72.76% vs 65.56%, $t\approx$16.4, $p<$0.001), with non-overlapping ranges (CKA-only: 72.10–73.09%, Standard: 65.13–65.43%).

M Cross-Sparsity Validation

Table 12: Cross-sparsity validation on DeepSeek-R1-Distill-Qwen-7B. GSM8K accuracy (N=1,319) after SFT warmup and GRPO recovery. 15% and 20% GRPO values are mean over 3 seeds (see Tables 2, 11). ‡Single seed. Mean $|\Delta| \leq 1$ pp across all conditions.

Sparsity	Method	SFT (%)	GRPO (%)	Δ (pp)
15%	CKA-only	72.18	**72.76**±**0.57**	+0.58
	Standard	66.49	65.56±0.50	−0.93
20%	CKA-only	63.91	64.39	+0.48
	Standard	59.29	59.84	+0.55
25%‡	CKA-only	60.12	59.59	−0.53
	Standard	50.87	51.86	+0.99
30%	All	Collapse (≤4%; Table 10)		

At 15% sparsity (3-seed mean), the CKA-only advantage widens to +7.20 pp (72.76% vs 65.56%), compared to +4.55 pp at 20%. CKA-only at 15% (72.76%) represents the highest mean accuracy across all experimental conditions, retaining 80.1% of original dense model performance (90.83%). At 25% sparsity (single seed), Standard's SFT accuracy decline accelerates (-7.20 pp from 15%→20%, -8.42 pp from 20%→25%), while CKA-only's decline decelerates (-8.27 pp then -3.79 pp). Reward-accuracy decoupling is preserved ($|\Delta| \leq 1$ pp for both methods) despite Standard's near-zero peak reward (0.008).

N Cross-Scale Validation

Table 13: Cross-scale validation at 20% sparsity. 7B: DeepSeek-R1-Distill-Qwen-7B (GSM8K mean over 3 seeds for CKA-only and Standard; see Table 2). 1.5B: DeepSeek-R1-Distill-Qwen-1.5B (single seed). GSM8K accuracy (N=1,319). 7B reward = peak across 100 GRPO steps (cf. Table 1); 1.5B reward = mean across training steps.

Method	7B		1.5B	
	Peak Rew.	GSM8K (%)	Rew.	GSM8K (%)
CKA-only	**0.354**	**64.39**	**0.238**	**35.48**
Standard	0.081	59.84	0.084	22.29
RASP (α=1.0)	0.061	60.88	0.099	26.16
Random	0.190	–	–	–

On DeepSeek-R1-Distill-Qwen-1.5B at 20% sparsity, the three-method ordering is preserved (CKA-only 35.48% $\gg$ RASP 26.16% $>$ Standard 22.29%), with reward-accuracy decoupling confirmed at this scale ($|\Delta| < 1$ pp: CKA-only -0.15, Standard -0.61 pp).

O Cross-Model Validation within Qwen2.5 Family

Table 14: Cross-model validation on Qwen2.5-Math-7B (Yang et al., 2024a) (same Qwen2.5 architecture family) at 20% structured sparsity. Accuracy evaluated on GSM8K (N=200). Post-SFT reports accuracy ± binomial standard error (N=200). EOS rate: fraction of generations that produce a valid end-of-sequence token.

Method	Post-Prune (%)	Post-SFT (%)	EOS Rate (%)
CKA-only	1.0	**51.50 ± 3.53**	27.0
Standard	18.5	21.00 ± 2.88	36.5

CKA-only pruning causes near-total initial damage on Qwen2.5-Math-7B (1.0% post-prune accuracy vs Standard's 18.5%), yet SFT recovers CKA-only to 51.50% while Standard remains at 21.00%. This post-prune accuracy reversal provides strong evidence that recovery capacity is shaped by preserved network structure rather than starting accuracy. EOS rates remain below 40% for both methods, indicating that SFT warmup is less effective on this architecture; GRPO validation is deferred to future work.[2]

[2]Preliminary GRPO evaluation yields 48.60% (243/500), but 83.8% of responses were truncated at 2048 tokens, precluding direct comparison with SFT results.

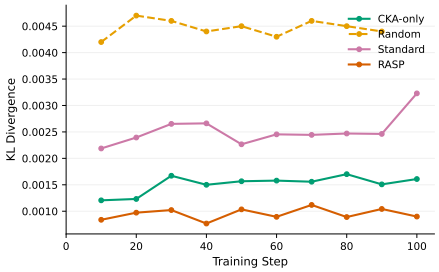


Figure 7: KL divergence between the policy and reference model during GRPO training. RASP maintains near-zero KL (<0.001), consistent with minimal policy movement during collapse. Random shows the highest KL (∼0.004–0.005), indicating more policy exploration. CKA-only and Standard fall between these extremes.

P Structural Overlap Analysis

CKA-only and Standard pruning produce markedly different removal patterns across the 28 layers of DeepSeek-R1-Distill-Qwen-7B at 20% sparsity. CKA-only concentrates pruning in early layers (L0–9 account for 72.5% of removed MLP groups), preserving nearly all structures in later layers (L16–27). Standard distributes removals more uniformly, pruning substantial numbers of groups in late layers (L16–18, L22, L25–26).

The methods share only partial overlap in which structures they remove. 65 MLP groups are pruned exclusively by CKA-only (retained by Standard), concentrated in L6–9 and L27; these are structures that gradient importance scores as important but CKA identifies as redundant with neighboring groups. Conversely, 46+ groups are pruned exclusively by Standard (retained by CKA-only), concentrated in L16–18, L22, and L25–26; these are structures that gradient importance scores as unimportant but CKA identifies as representationally unique.

This pattern is consistent with the layer-wise CKA similarity analysis (Figure 5): CKA-only's aggressive early-layer pruning disrupts dense representations (low diagonal CKA, μ=0.62) but preserves diverse late-layer pathways, while Standard's uniform pruning maintains dense fidelity (μ=0.90) at the cost of removing unique computational pathways in later layers.

Q KL Divergence During GRPO Training

R Depth-First Pruning Baseline

As a sanity check, we test whether CKA-only's advantage reflects mere layer-depth bias by constructing a depth-first baseline: synthetic importance scores force removal of late layers first (L18–27: 0.001, L10–17: 0.5, L0–9: 1.0), then standard pruning at 20% sparsity. Both SFT (∼10%) and GRPO (∼8%) accuracy are near chance (early-stopped at N≈50–88 samples due to conclusive results), confirming that naive depth ordering cannot substitute for criterion-based structure selection. This baseline operates at layer-level granularity (removing entire transformer blocks), unlike CKA-only/Standard/RASP which prune at MLP channel group level; the granularity difference limits direct comparison with our main results and precludes using this baseline as evidence for the recovery capacity spectrum.

S Per-Layer Capacity Floor Ablation

Table 15: Effect of per-layer capacity floor τ on CKA-only pruning at 20% sparsity. GSM8K accuracy (N=1,319) after GRPO recovery (100 steps). Default τ=0.4.

Floor τ	GRPO Acc. (%)	Δ vs τ=0.4 (pp)
0.3	65.05	+0.68
0.4 (default)	64.37	—
0.5	61.71	−2.66
0.6	62.02	−2.35

The per-layer capacity floor τ sets the minimum fraction of MLP channel groups retained in each layer, preventing over-pruning of any single layer regardless of its importance score. We ablate $\tau \in \{0.3, 0.4, 0.5, 0.6\}$ using CKA-only pruning at 20% sparsity. Lowering τ to 0.3 slightly improves accuracy (+0.68 pp), while raising it to 0.5 or 0.6 causes a notable drop (−2.66 and −2.35 pp respectively; the difference between 0.5 and 0.6 is not significant at 0.31 pp $\ll$ stderr $\approx$ 1.3 pp). Over-constraining per-layer retention hurts global pruning quality by forcing removal of important structures in unconstrained layers.

T TOST Equivalence Testing

To formally assess whether SFT and GRPO produce equivalent GSM8K accuracy, we apply Two One-Sided Tests (TOST) on N=1,319 paired per-example outcomes for CKA-only and Standard at 20% sparsity across multiple GRPO seeds. For each pair, $d_i = \mathbb{1}[\text{SFT correct}]_i - \mathbb{1}[\text{GRPO correct}]_i$, and we test H_0: $|\bar{d}| \geq \varepsilon$ against H_A: $|\bar{d}| < \varepsilon$.

Table 16: TOST equivalence results (ε=2 pp). All observed $|\Delta|$<1 pp.

Criterion	GRPO seed	Δ (pp)	p_{TOST}
CKA-only	seed 42	−0.45	0.008
CKA-only	seed 123	−0.30	0.005
CKA-only	verify	−0.15	0.001
CKA-only	seed 456	−0.68	0.023
Standard	seed 123	−0.68	0.020
Standard	seed 456	−0.99	0.069
Standard	fair_v2	−0.23	0.010
Standard	ext 300	−0.08	0.007

At ε=2 pp, 7/8 comparisons reject the null (p<0.05); the remaining pair (Standard vs. seed 456, Δ=−0.99 pp) is borderline (p=0.069). At ε=3 pp, all 8/8 are significant (p<0.002). The ε=1 pp margin lacks statistical power with binary outcomes at N=1,319.

U Use of AI Assistants

We used Claude (Anthropic) as an AI coding and writing assistant throughout this work. Specifically:

- **Code:** drafting experiment scripts, data-processing pipelines, and analysis utilities; debugging and refactoring.
- **Writing:** drafting and revising paper sections, polishing prose, and reformatting LaTeX.
- **Literature:** summarizing related papers to inform the related-work section.